INTERNSHIP TASK ANSWER 27/10/26 - Sarika Pansare

Section A – AI Companies & Latest Models

1. **ChatGPT** → Developed by **OpenAI**
2. **Claude** → Developed by **Anthropic**
3. **Gemini** → Developed by **Google DeepMind**
4. **Llama** → Developed by **Meta AI**
5. **Grok** → Developed by **xAI (Elon Musk)**
6. **Mistral AI HQ** → **France (Paris)**
7. **OpenAI's flagship GPT-6 frontier model** → **GPT-6 Astra**
8. **OpenAI's lower-cost near-Astra model** → **GPT-6.1 Sol**
9. **Anthropic's latest Haiku (Oct 2026)** → **Claude Haiku 5.5**
10. **Google's latest flagship Gemini (Oct 2026)** → **Gemini 4 Argon**

Section B – Generative AI & LLM Concepts

11. **LLM** → Large Language Model
12. **Generative AI** → AI that creates new content (text, image, audio, video)
13. **Prompt** → Instruction/input given to AI
14. **System Prompt** → Hidden guiding instruction that sets AI's behavior
15. **AI Hallucination** → False or fabricated output by AI
16. **Token** → Small unit of text processed by LLM
17. **Context Window** → Maximum tokens an LLM can remember at once
18. **Inference** → AI generating predictions/outputs from input
19. **Fine-tuning** → Training a pre-trained model with specialized data
20. **Multimodal AI** → AI that handles multiple data types (text + image + audio)

Section C – Agents, RAG & Modern AI Development

21. **AI Agent** → Autonomous AI system that performs tasks using tools/data
22. **Difference (Chatbot vs Agent)** → Chatbot = simple Q&A; Agent = proactive, tool-using, task-oriented
23. **RAG** → Retrieval-Augmented Generation
24. **Why RAG with LLMs?** → To improve accuracy by retrieving external knowledge
25. **Embedding** → Numeric representation of text meaning for search/comparison
26. **Vector Database** → Stores embeddings for fast semantic search
27. **Tool Calling** → Allows AI to use external tools/functions during reasoning
28. **MCP** → Model Context Protocol
29. **API Key** → Secure credential to access an AI service
30. **Agentic Workflow** → AI agents collaborating across tasks/tools in pipelines

Section D – AI Image, Video, Audio & Famous Tools

31. **Midjourney** → Famous for **image generation (art/visuals)**
32. **Sora** → **Text-to-video generation** (OpenAI)
33. **OpenAI's image model family** → **DALL·E**
34. **Text-to-Image** → Generating images from text prompts
35. **Text-to-Video** → Generating video from text prompts
36. **TTS** → Text-to-Speech
37. **STT** → Speech-to-Text
38. **OCR** → Optical Character Recognition (extracts text from images)
39. **Vision-Language Model** → AI that links visual input with language understanding
40. **Voice Cloning** → AI replicating a person's voice from samples

Section E – Coding AI, Hardware & Practical Awareness

41. **GitHub Copilot** → Used for **AI coding assistance**

42. **Cursor** → Used for **AI-powered coding IDE/workflows**
43. **AI Coding Assistant** → AI tool that helps write/debug code
44. **Hardware for training LLMs** → **GPU (Graphics Processing Unit)**
45. **GPU** → Graphics Processing Unit
46. **NPU** → Neural Processing Unit
47. **VRAM** → Stores model weights & activations during AI workloads
48. **Running AI locally** → Executing models on personal hardware instead of cloud
49. **Open-weight AI model** → Model with publicly available weights for use/modification
50. **Model Distillation** → Compressing a large model into a smaller, efficient one

Section F – Latest AI Awareness & 2026 News

51. **OpenAI model family (Astra, Sol, Luna)** → **GPT-6 family**
52. **Most demanding frontier GPT-6 model** → **GPT-6 Astra**
53. **Claude 5.5 model for complex coding/knowledge work** → **Claude Opus 5.5**
54. **Claude 5.5 model for fast, cost-sensitive work** → **Claude Haiku 5.5**
55. **Google Gemini 4 flagship** → **Gemini 4 Argon**
56. **OpenAI image generation family (2026)** → **DALL·E 4**
57. **OpenAI realtime voice model family** → **OpenAI Realtime Voice (Whisper successor)**
58. **Claude Opus, Sonnet, Haiku families** → Developed by **Anthropic**
59. **AI safety risk (malicious hidden instructions)** → **Prompt Injection**
60. **Why verify AI outputs in critical fields?** → Because AI can hallucinate; **human verification ensures accuracy, safety, and reliability**